

Document Title: Test Coverage – Course and Tutor Registration Feature

Application: MindBridge AI Tutor

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Version: 1.0

Feature Scope: Multi-step course and tutor registration workflow, including: Session browsing, cart management, and mock payment checkout.

Test Coverage Table:

Test Type	Description	Environment	Environment Justification	Pass/Fail Criteria
Functional Testing	Does the registration workflow complete correctly from start to finish?	QA Environment (localhost:8081)	Functional testing is assigned to the QA environment because it requires a fully operational application and a populated test database but does not require real student PII. The QA environment's scrubbed database provides realistic test data without exposing sensitive information, making it the appropriate and safe choice for this test type.	Test passes if workflow completes successfully with no errors from start to finish. If there are any errors or unexpected results, test is failed.
Negative / Error Handling Testing	What happens when invalid or unexpected input is entered? (Consider the known HTTP 500 defect triggered by special characters in the payment field.)	QA Environment (localhost:8081)	This test is assigned to the QA environment because it needs a semi-realistic environment, but there is no need for real student data. There is a need for us to use Browser Network tab in order to identify correctly the defect and resulting HTTP 500 response. Putting this test in Staging environment also increases negative brand risk during UAT testing.	Test passes if rather than reaching an HTTP 500 response the application validates the payment field for only approved characters. If there is no validation, user messaging, or we do receive a response error code, the test is failed.
Regression Testing	After a bug fix or update, does the registration workflow still work as expected?	QA Environment (localhost:8081)	Regression testing can be assigned to the QA environment since it requires a fully operational application and a populated test database but not real information. We need to ensure that the entire workflow continues to operate in a normal environment.	Test passes if all previous changes to the workflow continue to work as required. If our change has caused any issues with previous expected results, the test fails.

Usability Testing	Can a real student navigate the multi-step registration process intuitively?	Staging Environment (localhost:8082)	Usability testing is UAT allowing a real student the ability to go through the workflow. They need to be able to accomplish their everyday tasks with an environment as close to real-world as possible. QA environment wouldn't be sufficiently structured for this test.	Test passes if real student can accomplish real-world tasks with no interruption or error. Real students will complete a survey of acceptance and note any issues found. If no issues are found, tests pass, but if issues are found test fails.
Responsive Design Testing	Does the registration workflow display and function correctly on smaller screen sizes? (Consider the known navigation overlap defect at viewports below 400px.)	QA Environment (localhost:8081)	This test is assigned to the QA environment because it needs a semi-realistic environment, but there is no need for real student data. Testers need to observe only the UI with the smaller screen size to fully test viewport defect. Putting this test in Staging environment also negative brand increases risk during UAT testing.	Tests pass if workflow display continues to function correctly after various screen size changes. Changes will include various viewports below 400px. If there are any overlap or other defects in display, the tests fail.

Known Defects Affecting This Feature:

Defect ID: DEF-001

Description: Entering a special character (for example, '@') into the payment field bypasses front-end validation and causes the application to return an HTTP 500 Internal Server Error.

Visible In: Browser Network tab

Recommended Test Environment: QA (localhost:8081)

Reason: This defect can be safely reproduced and documented in the QA environment using anonymized test payment data. Testing in QA eliminates any risk to real user accounts or financial systems while allowing developers to investigate the issue.

Defect ID: DEF-002

Description: When the browser viewport is reduced below 400px width (mobile emulation), the primary navigation menu overlaps the main content area, making it impossible to interact with.

Visible In: User Interface – Navigation

Recommended Test Environment: QA (localhost:8081)

Reason: This test is assigned to the QA environment because it needs a semi-realistic environment, but there is no need for real student data. Testers need to observe only the UI with the smaller screen size to fully test viewport defect. By putting this test in Staging environment negative brand increases risk during UAT testing.